

ARIF SYRAJ

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EDUCATION

University of San Francisco

Master of Science, Computer Science

San Francisco, CA

August 2023 – December 2025

Relevant Coursework: Algorithms, Principles of Software Development, Foundations of A.I., Systems Programming, Software Development Lifecycle, Cloud Computing

University of Illinois at Urbana-Champaign

Bachelor of Science, Mechanical Engineering

Champaign, IL

August 2016 - May 2021

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, HTML5, CSS3, SQL

General: Finetuning, Prompt Engineering, Agile, Data Scraping, Shell Scripting, Containerization, Multithreading, CI/CD, Automated Testing, Cloud Deployment, Network Protocols

Frameworks/Tools: Linux, AWS, GCP, Docker, ROS, pytorch, tensorflow, MongoDB, React, Node.js, Express, JUnit

EXPERIENCE

Teaching and Research Assistant - AI For Redistricting

University of San Francisco

January 2025 – May 2025

San Francisco, CA

Software Engineer

EarthSense, Inc.

March 2021 – July 2022

Champaign, IL

- Designed an algorithm to estimate plant height using noisy LiDAR data, cutting error by >50% across datasets over the previous algorithm and beating accuracy of manual measurements.
- Revamped robot autonomy by designing and finetuning a crash detection algorithm fusing LiDAR and odometry data collected from various sensors, reducing overall errors in crash detection by 75%.
- Implemented automatic waypoint generation for robot navigation, saving customers hours during setup time.
- Standardized testing methodologies and created shell scripts to catch system bugs before deployment.

PROJECTS

Duets AI

August - December 2025

- Led a 3-person industry-sponsored team to ship AI, backend and frontend features for an AI-augmented language-learning platform using OpenAI, Python, Django and Azure.
- Cut company-wide LLM spend by 80% by switching models, prompt engineering, and backend changes.
- Lowered server and storage costs by 30% by reworking frontend and backend logic, and migrating tens of thousands of hot data entries from cold storage to Supabase.
- Built meeting transcript-based exercise generation; added checks, retries, and fallbacks to make it 99.9% reliable.
- Coordinated scope and delivery with sponsors and two student teams; enforced issue/PR tracking and review standards to keep releases moving efficiently.

Highly Available Cloud Application Deployment

October 2024

- Architected and deployed a highly available, secure application on AWS using ECR, EC2, ALB, VPC, and ASGs.
- Configured an ALB to distribute traffic across EC2 instances in private subnets across multiple AZ's with launch templates and auto-scaling groups, enhancing scalability, security and availability.
- Improved operational security by launching bastion hosts across availability zones using ASGs for highly available, secure SSH access to EC2 instances.
- Set up TLS encryption using ACM and Route 53 for domain management and HTTPS-encrypted communication.
- Deployed a containerized application orchestrated with ECS.

LLM's To Detect Chart Misinformation

August 2024

- Finetuned a multimodal LLM (LLaVA) to detect visual misinformation in bar charts with LoRA and 2.3k examples.
- Created a custom finetuning script and explored parameters to prevent overfitting and minimize GPU loads.
- Resulted in a model that can identify visually manipulated bar charts 90% of the time.

AWARDS/PUBLICATIONS

Author - "States of Disarray: Cleaning Data for Gerrymandering Analysis"

March 2025

University of San Francisco Computer Science Scholarship - Merit Based Award

May 2023

EnterpriseWorks Incubator Top Summer Intern Award - "Best Technical Innovation"

July 2021